Technical Report: Backend Data Engineering & Predictive Engine

Project Title: Delivery Risk Prediction and Geographical Scope Dashboard– APL Logistics

(Proactive Logistics Logic & Predictive Modeling)

Industry Partner: APL Logistics

Organization: Unified Mentor Pvt. Ltd. (Internship 2026)

Tool Used: Python 3.10.12 (Google Colab)

Environment: Cloud-based Jupyter Notebook

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1. Statistical Analysis (Descriptive Summary)

The analysis of the distribution and central tendency for 'Days for shipping (real)' and 'Days for shipment (scheduled)' is as follows:

Metric	Days for shipping (real)	Days for shipment (scheduled)
Mean (Average)	3.50 Days	2.93 Days
Median (Middle)	3.00 Days	4.00 Days
Mode (Frequent)	2.00 Days	4.00 Days

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2. KPI Formulas

KPI	Formula
Delivery Gap	Days for shipping (real) - Days for shipment (scheduled)
On-Time Delivery Rate	(Total On-Time Orders / Total Orders) * 100

Average Delivery Delay	Sum of (Delay Days) / Total Number of Late Orders
Late Delivery Risk Ratio	(Total Late Orders / Total Orders) * 100

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3. Delivery Gap Findings & Predictive Modeling

- **Delivery Gap Analysis:** The data indicates that a significant portion of orders is delivered beyond the planned schedule. While most orders are delivered within 2-3 days, a substantial volume extends to 5-6 days, exceeding the planned timeline.
- **Performance Metrics:**
  - **On-Time Delivery Rate:** 42.72%
  - **Average Delivery Delay:** 1.62 Days
  - **Late Delivery Risk Ratio:** 57.28%
- **Predictive Modeling:** A Logistic Regression model was implemented to predict late delivery risks based on features like 'Shipping Mode' and 'Order Region'.
- **Technical Implementation:**
  - **Model Training (model.fit):** The algorithm learns historical patterns and fits data points to the sigmoid curve
$$P = \frac{1}{1 + e^{-z}}$$
  - **Risk Prediction (model.predict_proba):** This function converts input data into a probability range of 0 to 100%. The [:, 1] parameter is specifically utilized to extract the probability of a 'Late Delivery' (1) event.

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4. Standardization

- **Geographical Data:** Data consistency in Order City, Order Country, and Market columns was achieved using **VLOOKUP** in Excel. This process identified and corrected special characters (accents), ensuring data integrity for categorization.

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5. Executive Summary

This project provides a robust engine for predictive logistics analytics. With 180,519 records analyzed, the findings highlight that **57.28%** of orders are at risk of late delivery.

Main Achievements & Future Scope:

- **Automated Risk Calculation:** Real-time risk generation via Logistic Regression.
- **Operational Efficiency:** Identification of bottlenecks through regional and shipping mode performance indices.
- **Data Integrity:** Standardized geographical data using normalization techniques.

- **Future Scope:** The model can be integrated into an **Automated Real-Time Alert System**. By setting a risk threshold (e.g., > 45%), the system can trigger instant alerts for high-risk orders, enabling proactive logistics intervention.

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6. Conclusion

This project establishes a strong foundation for identifying hidden inefficiencies in logistics operations. By combining systematic data cleaning with predictive modeling, APL Logistics can transition from reactive performance evaluation to proactive delay mitigation, ultimately driving operational excellence.